

HDOC DAY-35 - C PROGRAMMING

Variable-Length Square Array: Main Diagonal Operations and Diagonal Traversal

Name	Dhruv Pandey
SAP ID	590037660
Batch	15
Activity	HDOC Day-35

Problem Statement

Write a menu-driven C program to read a variable-length square integer array and perform the following operations using separate functions:

1. Determine whether all the elements on its main diagonal are distinct.
2. Find the sum of its main diagonal elements.
3. Display its elements using diagonal traversal, beginning at the top-left corner and traversing successive diagonals alternately in upward-right and downward-left directions.
4. Exit.

Algorithm

1. Start and read the order n of the square matrix.
2. Read all $n \times n$ integer elements into a variable-length array.
3. Display a menu repeatedly until the user chooses Exit.
4. For distinct diagonal elements, compare each $a[i][i]$ with every later diagonal element $a[j][j]$. If any pair is equal, the diagonal elements are not distinct.
5. For diagonal sum, add $a[i][i]$ for $i = 0$ to $n-1$.
6. For diagonal traversal, process diagonals by the value $s = \text{row} + \text{column}$. For even s , move upward-right; for odd s , move downward-left.
7. Stop when Exit is selected.

Test Case Table

Test	Input / Matrix	Operation	Expected Result
1	3 x 3: 1 2 3 / 4 5 6 / 7 8 9	Distinct	1, 5, 9 are distinct
2	Same matrix	Sum	15
3	Same matrix	Traversal	1 2 4 7 5 3 6 8 9
4	3 x 3: 1 2 3 / 4 1 6 / 7 8 9	Distinct	Not distinct

C Program

```
#include <stdio.h>

void readMatrix(int n, int a[n][n]) {
    printf("Enter the matrix elements:\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
}

void checkDistinctDiagonal(int n, int a[n][n]) {
    int distinct = 1;
    printf("Main diagonal elements: ");
    for (int i = 0; i < n; i++)
        printf("%d ", a[i][i]);

    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i][i] == a[j][j]) {
                distinct = 0;
                break;
            }
        }
        if (!distinct) break;
    }

    if (distinct)
        printf("\nAll the main diagonal elements are distinct.\n");
    else
        printf("\nAll the main diagonal elements are not distinct.\n");
}

void diagonalSum(int n, int a[n][n]) {
    int sum = 0;
    for (int i = 0; i < n; i++)
        sum += a[i][i];
    printf("Sum of main diagonal elements: %d\n", sum);
}

void diagonalTraversal(int n, int a[n][n]) {
    printf("Diagonal traversal: ");

    for (int s = 0; s <= 2 * (n - 1); s++) {
        if (s % 2 == 0) {
            int i = (s < n) ? s : n - 1;
            int j = s - i;
            while (i >= 0 && j < n) {
                printf("%d ", a[i][j]);
                i--;
                j++;
            }
        } else {
            int j = (s < n) ? s : n - 1;
            int i = s - j;
            while (j >= 0 && i < n) {
                printf("%d ", a[i][j]);
                i++;
                j--;
            }
        }
    }
    printf("\n");
}

int main() {
    int n, choice;

    printf("Enter the order of the square matrix: ");
    scanf("%d", &n);

    int a[n][n];
    readMatrix(n, a);

    do {
        printf("\n--- MENU ---\n");
        printf("1. Check distinct main diagonal elements\n");
    }
```

```

printf("2. Sum of main diagonal elements\n");
printf("3. Diagonal traversal\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        checkDistinctDiagonal(n, a);
        break;
    case 2:
        diagonalSum(n, a);
        break;
    case 3:
        diagonalTraversal(n, a);
        break;
    case 4:
        printf("Exiting program.\n");
        break;
    default:
        printf("Invalid choice. Please try again.\n");
}
} while (choice != 4);

return 0;
}

```

Sample Execution

```
Enter the order of the square matrix: 3
Enter the matrix elements:
1 2 3
4 5 6
7 8 9

--- MENU ---
1. Check distinct main diagonal elements
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 1
Main diagonal elements: 1 5 9
All the main diagonal elements are distinct.

Enter your choice: 2
Sum of main diagonal elements: 15

Enter your choice: 3
Diagonal traversal: 1 2 4 7 5 3 6 8 9

Enter your choice: 4
Exiting program.
```

Result

The menu-driven C program successfully checks whether the main diagonal elements are distinct, calculates their sum, and displays the square matrix in the required alternating diagonal traversal order.

Execution

HDOC Day-35 - C Programming

```
Last login: Tue Sep 29 09:39:29 on console
dhruvpandey@DHruVs-MacBook-Air ~ % nano day35.c
dhruvpandey@DHruVs-MacBook-Air ~ % clang day35.c -oday35
dhruvpandey@DHruVs-MacBook-Air ~ % ./day35
Enter the order of square matrix: 1
Enter the matrix elements:
2

----- MENU -----
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
Enter your choice: 1

Main diagonal elements: 2
All the main diagonal elements are distinct.

----- MENU -----
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
Enter your choice: 2

Sum of main diagonal elements: 2

----- MENU -----
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
Enter your choice: 3

Diagonal traversal: 2

----- MENU -----
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
Enter your choice: 4
Exiting program...
dhruvpandey@DHruVs-MacBook-Air ~ %
```

Program compilation and execution in Terminal